Circular Apertures and Resolution

> SYNOPSIS
>
> Every lens, mirror and pupil is a hole, and a hole diffracts. So no instrument ever makes a point image of a point object — it makes a small blur. When two objects' blurs overlap, the instrument cannot tell them apart, and **that limit is set by physics, not by workmanship**.

BEFORE YOU START — PREDICT

You are told a telescope mirror has been ground and polished **perfectly** — no flaw anywhere. Can it now resolve two stars that are arbitrarily close together? Answer yes or no, and say what decides it.

PERFORMANCE EXPECTATIONS

1. **Explain** why a sharp-edged shadow is evidence *against* diffraction, and a fuzzy one evidence for it.
2. **State** the Rayleigh criterion $\theta_m = 1.22\lambda/D$ and say what each symbol is.
3. **Compute** the resolving angle of an instrument from its aperture and the wavelength.
4. **Convert** a resolving angle into a physical separation at a known distance, using $s = r\theta$.

Circular aperture and resolution

If there were **no diffraction**, light through a round hole would throw a shadow with a **sharp edge** — a clean disc, not "fuzzy". It does not. The edge is blurred, and the blurring is diffraction at the rim of the aperture.

Now put **two** point sources in front of that aperture. Each makes its own blur. If the blurs are far apart you see two objects; if they overlap enough, you see one.

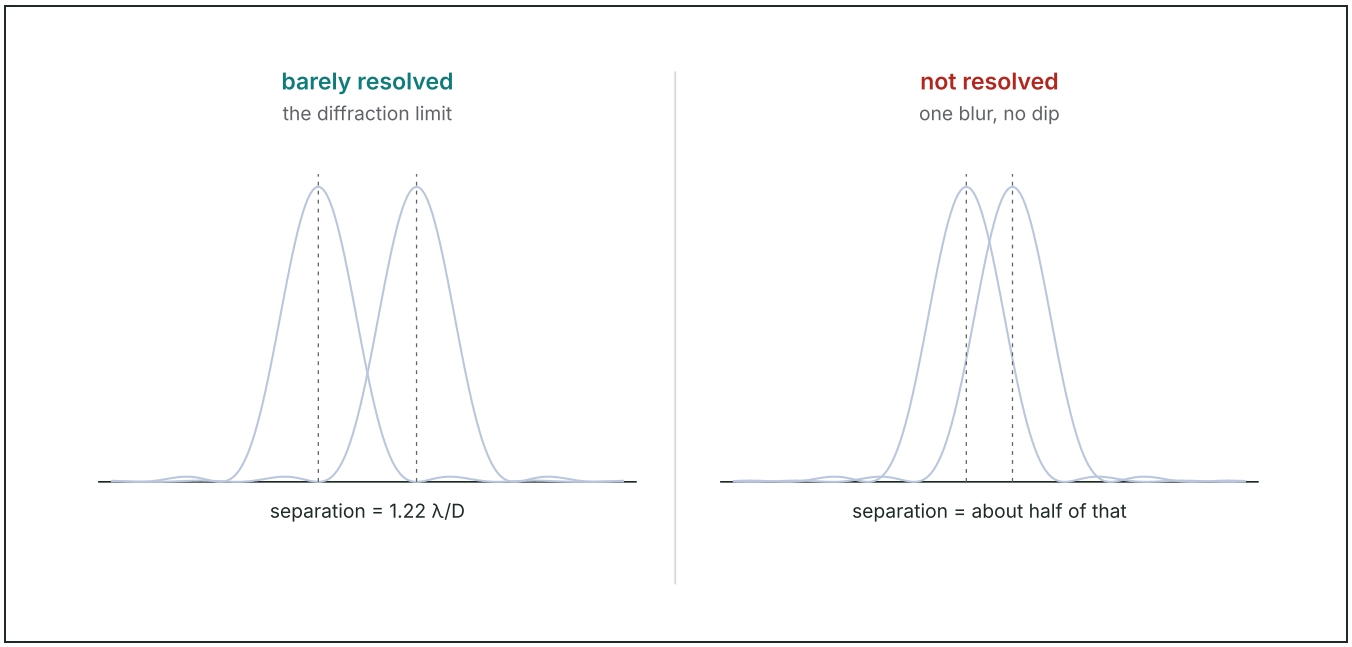


Figure 1. Each curve is a real Airy pattern — $(2J_1(x)/x)^2$ — and the pale curves are the two sources taken separately. On the left they sit exactly **one first-zero apart**, and the dip in their sum is the actual dip. Push them closer and it fills in.

Your job on Figure 1. Each panel shows the two sources' patterns separately, in pale blue. Draw their **sum** in each — that is what the eye receives. Then decide which pair you could honestly call two objects, and say what you used to decide.

The dividing line is the angle at which the **peak of one pattern falls on the first dark ring of the other**. For a circular aperture that angle is

$$\theta_m = 1.22\,\lambda / D$$

where D is the diameter of the aperture. θ_m is **the angle for just about resolving** the two. Closer than that and they are **not resolved** — no amount of magnification recovers them, because the information is not in the light any more.

Worked example — the Hubble Space Telescope

(1) The Hubble mirror has D = **2.4 m**. At λ = **550 nm**, what is θ_m?

(2) Two stars sit in a galaxy **2 million light years** away. How far apart must they be for Hubble to see them as two?

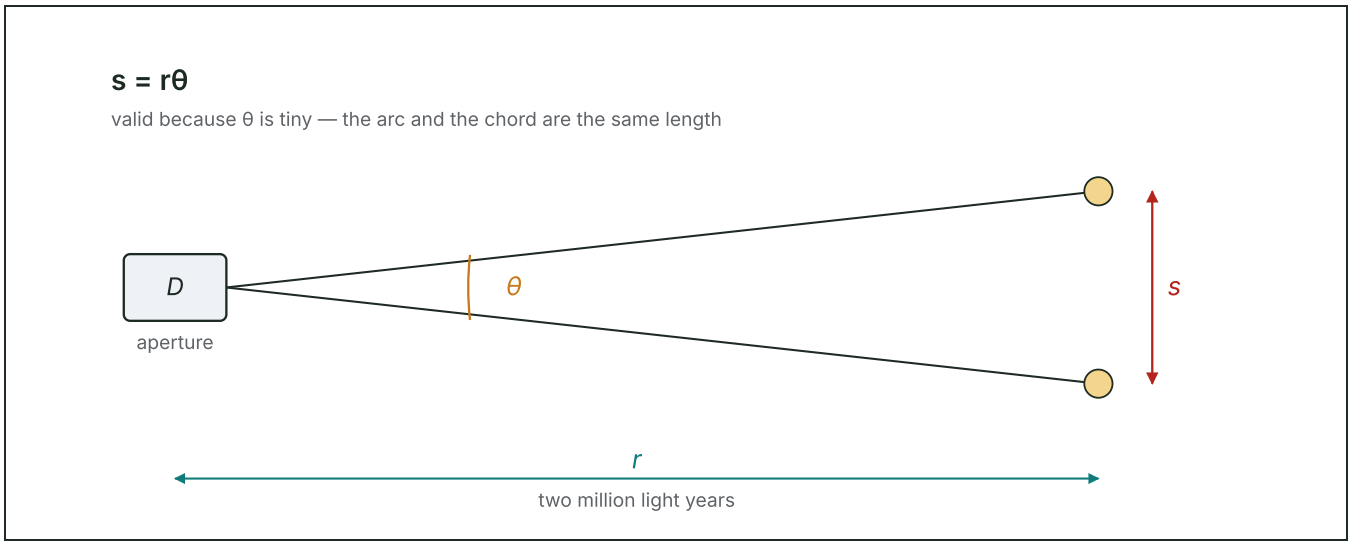


Figure 2. Once you have the smallest angle a telescope can resolve, the smallest *separation* it can resolve at a known distance is one line of geometry. **The answer scales with how far away the objects are**, which is why resolving a pair of stars in another galaxy is a far harder problem than resolving two headlights.

1. You have an angle and a distance. The rest is one line — but check what a light year is before you write it.

THE PUNCHLINE

$$\theta_m = 1.22\,\lambda / D$$

Back to the prediction: **no**. A perfect mirror still cannot resolve two arbitrarily close stars, because the limit is set by the *wavelength of the light* and the *size of the hole*, not by the quality of the glass. To do better you need a bigger aperture or a shorter wavelength — which is exactly why telescopes keep getting larger.

Looking back

That closes Module 2. It started with light as a ray that travels in straight lines and bends at surfaces, and ends with a limit on what any instrument can ever see — one that exists only because light is a **wave**. Nothing in Notes 5 to 9 could have predicted it.

2. A telescope has an aperture of 0.50 m. At 500 nm, what is θ_m, in microradians?

_____ µrad

3. A mirror is ground perfectly, with no flaw at all. Can it resolve two arbitrarily close stars?

○ Yes — a perfect mirror has no resolution limit. ○ No — diffraction at the aperture sets a limit regardless of quality. ○ Yes, as long as enough magnification is used. ○ Only for wavelengths longer than the aperture.

4. To improve the resolution of a telescope, you should:

○ Increase the aperture, or observe at a shorter wavelength. ○ Decrease the aperture, or observe at a longer wavelength. ○ Increase the magnification of the eyepiece. ○ Move the telescope closer to the object.

Check Clear

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